

Supporting Information for
Stiff-PINN: Physics-Informed Neural Network for Stiff Chemical Kinetics

Weiqi Ji ^{a†}, Weilun Qiu ^{b†}, Zhiyu Shi ^{b†}, Shaowu Pan ^c, Sili Deng ^{a*}

^a *Department of Mechanical Engineering, Massachusetts Institute of Technology,*
Cambridge, MA 02139, USA

^b *College of Engineering, Peking University, Beijing 100871, China*

^c *Department of Aerospace Engineering, University of Michigan, Ann Arbor, MI 48109, USA*

[†] These authors contributed equally to this work.

^{*} Corresponding Author: Sili Deng (silideng@mit.edu)

A. Details of the pollution model

The reaction scheme of the POLLU problem is as follows.

Table A.1. The reaction scheme of the POLLU problem.

No.	Reaction	No.	Reaction
1	$NO_2 \rightarrow NO + O_3P$	14	$NO_2 + OH \rightarrow HNO_3$
2	$NO + O_3 \rightarrow NO_2$	15	$O_3P \rightarrow O_3$
3	$HO_2 + NO \rightarrow NO_2 + OH$	16	$O_3 \rightarrow O_1D$
4	$HCHO \rightarrow 2HO_2 + CO$	17	$O_3 \rightarrow O_3P$
5	$HCHO \rightarrow CO$	18	$O_1D \rightarrow 2OH$
6	$HCHO + OH \rightarrow HO_2 + CO$	19	$O_1D \rightarrow O_3P$
7	$ALD \rightarrow MEO_2 + HO_2 + CO$	20	$SO_2 + OH \rightarrow SO_4 + HO_2$
8	$ALD + OH \rightarrow C_2O_3$	21	$NO_3 \rightarrow NO$
9	$C_2O_3 + NO \rightarrow NO_2 + MEO_2 + CO_2$	22	$NO_3 \rightarrow NO_2 + O_3P$
10	$C_2O_3 + NO_2 \rightarrow PAN$	23	$NO_2 + O_3 \rightarrow NO_3$
11	$PAN \rightarrow C_2O_3 + NO_2$	24	$NO_3 + NO_2 \rightarrow N_2O_5$
12	$MEO_2 + NO \rightarrow CH_3O + NO_2$	25	$N_2O_5 \rightarrow NO_3 + NO_2$
13	$CH_3O \rightarrow HCHO + HO_2$		

The net production rate of each species is given in Table A.2.

Table A.2. The net production rate of each species.

No.	Net production rate	No.	Net production rate
1	$-\sum_{j \in \{1, 10, 14, 23, 24\}} r_j + \sum_{j \in \{2, 3, 9, 11, 12, 22, 25\}} r_j$	11	$-r_9 - r_{10} + r_8 + r_{11}$
2	$-r_2 - r_3 - r_9 - r_{12} + r_1 + r_{21}$	12	r_9
3	$-r_{15} + r_1 + r_{17} + r_{19} + r_{22}$	13	$-r_{11} + r_{10}$
4	$-r_2 - r_{16} - r_{17} - r_{23} + r_{15}$	14	$-r_{13} + r_{12}$
5	$-r_3 + 2r_4 + r_6 + r_7 + r_{13} + r_{20}$	15	r_{14}
6	$-r_6 - r_8 - r_{14} - r_{20} + r_3 + 2r_{18}$	16	$-r_{18} - r_{19} + r_{16}$
7	$-r_4 - r_5 - r_6 + r_{13}$	17	$-r_{20}$
8	$r_4 + r_5 + r_6 + r_7$	18	r_{20}
9	$-r_7 - r_8$	19	$-r_{21} - r_{22} - r_{24} + r_{23} + r_{25}$
10	$-r_{12} + r_7 + r_9$	20	$-r_{25} + r_{24}$

where the r_i are auxiliary variables as follows.

Table A.3. The rates of each reaction.

r	k	r	k
$r_1 = k_1 y_1$	$k_1 = 0.350$	$r_{14} = k_{14} y_1 y_6$	$k_{14} = 0.163\text{E}5$
$r_2 = k_2 y_2 y_4$	$k_2 = 0.266\text{E}2$	$r_{15} = k_{15} y_3$	$k_{15} = 0.480\text{E}7$
$r_3 = k_3 y_5 y_2$	$k_3 = 0.123\text{E}5$	$r_{16} = k_{16} y_4$	$k_{16} = 0.350\text{E} - 3$
$r_4 = k_4 y_7$	$k_4 = 0.860\text{E} - 3$	$r_{17} = k_{17} y_4$	$k_{17} = 0.175\text{E} - 1$
$r_5 = k_5 y_7$	$k_5 = 0.820\text{E} - 3$	$r_{18} = k_{18} y_{16}$	$k_{18} = 0.100\text{E}9$
$r_6 = k_6 y_7 y_6$	$k_6 = 0.150\text{E}5$	$r_{19} = k_{19} y_{16}$	$k_{19} = 0.444\text{E}12$
$r_7 = k_7 y_9$	$k_7 = 0.130\text{E} - 3$	$r_{20} = k_{20} y_{17} y_6$	$k_{20} = 0.124\text{E}4$
$r_8 = k_8 y_9 y_6$	$k_8 = 0.240\text{E}5$	$r_{21} = k_{21} y_{19}$	$k_{21} = 0.210\text{E}1$
$r_9 = k_9 y_{11} y_2$	$k_9 = 0.165\text{E}5$	$r_{22} = k_{22} y_{19}$	$k_{22} = 0.578\text{E}1$
$r_{10} = k_{10} y_{11} y_1$	$k_{10} = 0.900\text{E}4$	$r_{23} = k_{23} y_1 y_4$	$k_{23} = 0.474\text{E} - 1$
$r_{11} = k_{11} y_{13}$	$k_{11} = 0.220\text{E} - 1$	$r_{24} = k_{24} y_{19} y_1$	$k_{24} = 0.178\text{E}4$
$r_{12} = k_{12} y_{10} y_2$	$k_{12} = 0.120\text{E}5$	$r_{25} = k_{25} y_{20}$	$k_{25} = 0.312\text{E}1$
$r_{13} = k_{13} y_{14}$	$k_{13} = 0.188\text{E}1$		

The order of the species and the initial values of the concentrations of the species are given in Table A.4.

Table A.4. The initial values of the concentrations of the species y_0 .

Variable	Species	Initial value	Variable	Species	Initial value
y_1	$[NO_2]$	0	y_{11}	$[C_2O_3]$	0
y_2	$[NO]$	0.2	y_{12}	$[CO_2]$	0
y_3	$[O_3P]$	0	y_{13}	$[PAN]$	0
y_4	$[O_3]$	0.04	y_{14}	$[CH_3O]$	0
y_5	$[HO_2]$	0	y_{15}	$[HNO_3]$	0
y_6	$[OH]$	0	y_{16}	$[O_1D]$	0
y_7	$[HCHO]$	0.1	y_{17}	$[SO_2]$	0.007
y_8	$[CO]$	0.3	y_{18}	$[SO_4]$	0
y_9	$[ALD]$	0.01	y_{19}	$[NO_3]$	0
y_{10}	$[MEO_2]$	0	y_{20}	$[N_2O_5]$	0

Similar to the ROBER problem, the POLLU problem is also very stiff. For example, the maximum concentration of y_2 is higher than y_{16} by 16 orders of magnitude. We then applied QSSA to ten intermediate species of [3, 5, 6, 10, 11, 13, 14, 16, 19, 20], i.e., assuming that the net production rates of these species are zero:

$$-k_{15}y_3 + k_1y_1 + k_{17}y_4 + k_{19}y_{16} + k_{22}y_{19} = 0 \quad (A.1)$$

$$-k_3y_5y_2 + 2k_4y_7 + k_6y_7y_6 + k_7y_9 + k_{13}y_{14} + k_{20}y_{17}y_6 = 0 \quad (A.2)$$

$$-k_6y_7y_6 - k_8y_9y_6 - k_{14}y_1y_6 - k_{20}y_{17}y_6 + k_3y_5y_2 + 2k_{18}y_{16} = 0 \quad (A.3)$$

$$-k_{12}y_{10}y_2 + k_7y_9 + k_9y_{11}y_2 = 0 \quad (A.4)$$

$$-k_9y_{11}y_2 - k_{10}y_{11}y_1 + k_8y_9y_6 + k_{11}y_{13} = 0 \quad (A.5)$$

$$k_{10}y_{11}y_1 = k_{11}y_{13} \quad (A.6)$$

$$k_{12}y_{10}y_2 = k_{13}y_{14} \quad (A.7)$$

$$-k_{18}y_{16} - k_{19}y_{16} + k_{16}y_4 = 0 \quad (\text{A.8})$$

$$-k_{21}y_{19} - k_{22}y_{19} - k_{24}y_{19}y_1 + k_{23}y_1y_4 + k_{25}y_{20} = 0 \quad (\text{A.9})$$

$$k_{24}y_{19}y_1 = k_{25}y_{20} \quad (\text{A.10})$$

In the ten equations above, $[y_3, y_5, y_6, y_{10}, y_{11}, y_{13}, y_{14}, y_{16}, y_{19}, y_{20}]$ are unknown quantities that need to be solved, and we want to express them as functions of non-QSS species via algebraic equations.

From (A.8), we get

$$y_{16} = \frac{k_{16}y_4}{k_{18} + k_{19}} \quad (\text{A.11})$$

From (A.6), we get

$$y_{13} = \frac{k_{10}y_{11}y_1}{k_{11}} \quad (\text{A.12})$$

where y_{11} is an unknown quantity to be solved later.

From (A.5) and (A.6), we get

$$y_6 = \frac{k_9y_{11}y_2}{k_8y_9} \quad (\text{A.13})$$

From (A.4), we get

$$y_{10} = \frac{k_7y_9 + k_9y_{11}y_2}{k_{12}y_2} \quad (\text{A.14})$$

From (A.7) and (A.14), we get

$$y_{14} = \frac{k_7y_9 + k_9y_{11}y_2}{k_{13}} \quad (\text{A.15})$$

From (A.9) and (A.10), we get

$$y_{19} = \frac{k_{23}y_1y_4}{k_{21} + k_{22}} \quad (\text{A.16})$$

From (A.10) and (A.16), we get

$$y_{20} = \frac{k_{23}k_{24}y_1^2y_4}{(k_{21} + k_{22})k_{25}} \quad (\text{A.17})$$

From (A.3) and (A.11), we get

$$y_5 = \frac{(k_6 y_7 + k_8 y_9 + k_{14} y_1 + k_{20} y_{17}) y_6 - \frac{2k_{18} k_{16} y_4}{k_{18} + k_{19}}}{k_3 y_2} \quad (\text{A.18})$$

where y_6 is determined by (A.13).

From (A.1), (A.11), and (A.16), we get

$$y_3 = \frac{k_1 y_1 + k_{17} y_4 + \frac{k_{16} k_{19} y_4}{k_{18} + k_{19}} + \frac{k_{22} k_{23} y_1 y_4}{k_{21} + k_{22}}}{k_{15}} \quad (\text{A.19})$$

From (A.2), (A.13), (A.15), and (A.18), we get

$$y_{11} = \frac{\left(\frac{2k_{16} k_{18} y_4}{k_{18} + k_{19}} + 2k_4 y_7 + 2k_7 y_9 \right) k_8 y_9}{k_9 k_{14} y_1 y_2} \quad (\text{A.20})$$

After substituting (A.20) into (A.12), (A.13), (A.14), (A.15), we can get the algebraic expression of each variable to be solved.

It is worth noting that it is not appropriate to apply QSSA on species 18 although it is also a species with rapid concentration changes. This is because if we do that, in addition to the above equations of (A. 1)-(A.10), we will also get

$$k_{20} y_{17} y_6 = 0 \quad (\text{A.21})$$

which implied that

$$y_6 = 0 \quad (\text{A.22})$$

This formula contradicts the equations of (A.11)-(A.20). That is to say, the number of equations will be greater than the number of variables to be solved if we also apply QSSA to species y_{18} .

B. Regular-PINN for the POLLU problem

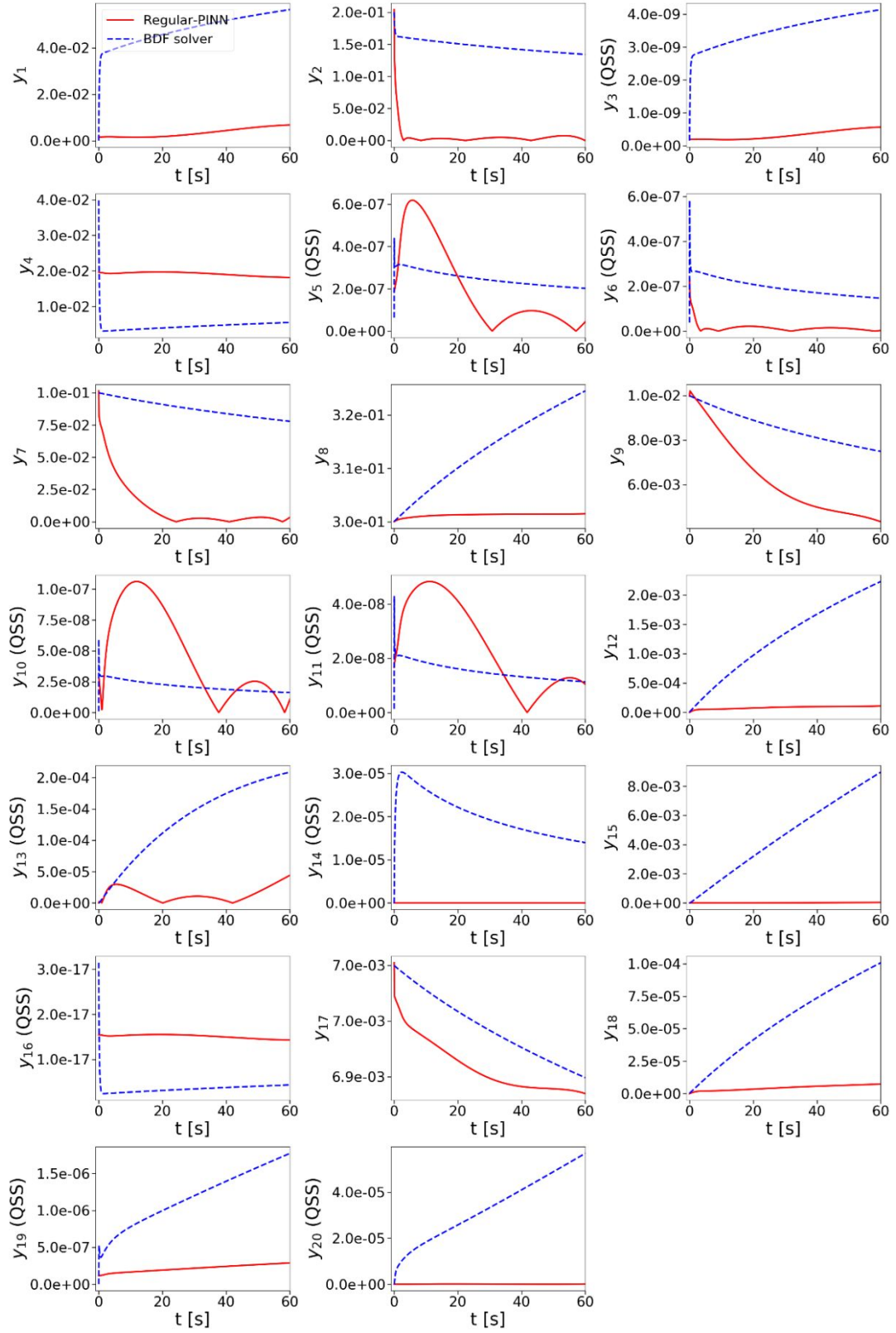


Figure S1. The training results of regular-PINN for the POLLU problem.